

Real-Time Interrupt (RTI) Module

This chapter describes the functionality of the real-time interrupt (RTI) module. The RTI is designed as an operating system timer to support a real time operating system (RTOS).

NOTE: This chapter describes a superset implementation of the RTI module that includes features and functionality related to DMA, and Timbase control. These features are dependent on the device-specific feature content. Consult your device-specific datasheet to determine the applicability of these features to your device being used.

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13.1 Overview

The real-time interrupt (RTI) module provides timer functionality for operating systems and for benchmarking code. The RTI module can incorporate several counters that define the timebases needed for scheduling in the operating system.

The timers also allow you to benchmark certain areas of code by reading the values of the counters at the beginning and the end of the desired code range and calculating the difference between the values.

13.1.1 Features

The RTI module has the following features:

- Two independent 64 bit counter blocks
- Four configurable compares for generating operating system ticks or DMA requests. Each event can be driven by either counter block 0 or counter block 1.
- Fast enabling/disabling of events
- Two time stamp (capture) functions for system or peripheral interrupts, one for each counter block
- Digital windowed watchdog

13.1.2 Industry Standard Compliance Statement

This module is specifically designed to fulfill the requirements for OSEK (**O**ffene **S**ysteme und deren **S**chnittstellen für die **E**lektronik im **K**raftfahrzeug, or Open Systems and the Corresponding Interfaces for Automotive Electronics) as well as OSEK/time-compliant operating systems, but is not limited to it.

13.2 Module Operation

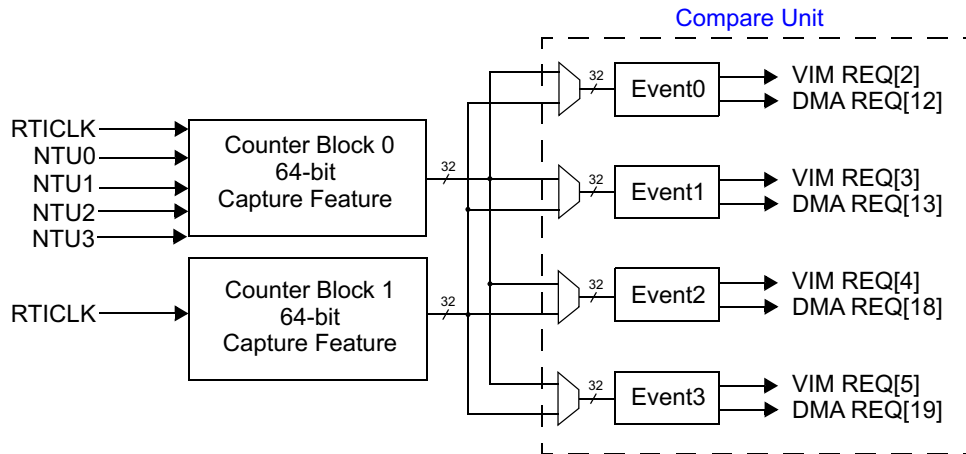
Figure 13-1 illustrates the high level block diagram of the RTI module.

The RTI module has two independent counter blocks for generating different timebases: counter block 0 and counter block 1.

A compare unit compares the counters with programmable values and generates four independent interrupt or DMA requests on compare matches. Each of the compare registers can be programmed to be compared to either counter block 0 or counter block 1.

The following sections describe the individual functions in more detail.

Figure 13-1. RTI Block Diagram



13.2.1 Counter Operation

Each counter block consists of the following (see Figure 13-2):

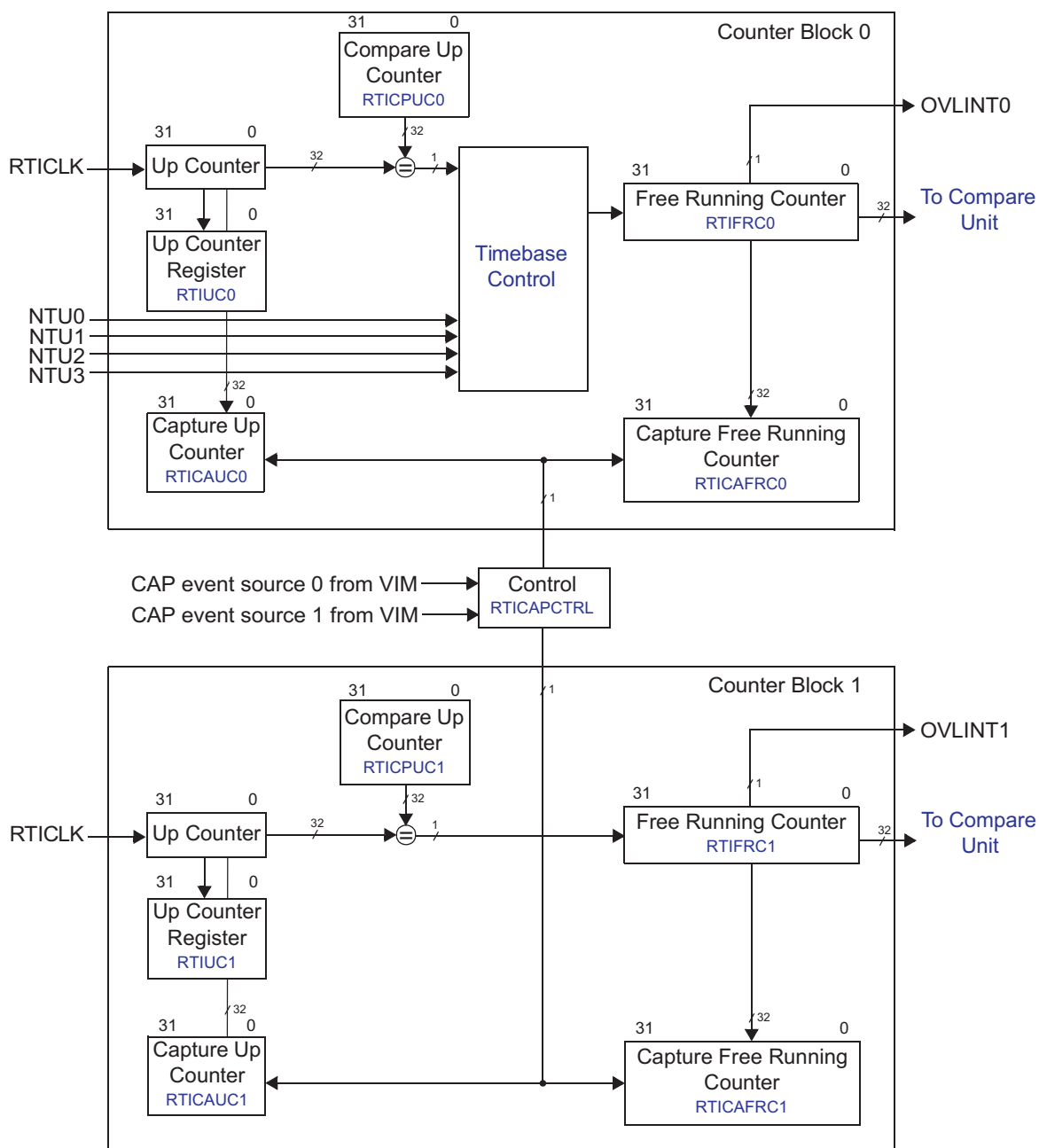
- One 32-bit prescale counter (RTIUC0 or RTIUC1)
- One 32-bit free running counter (RTIFRC0 or RTIFRC1)

The RTIUC0/1 is driven by the RTICLK and counts up until the compare value in the compare up counter register (RTICPUC0 or RTICPUC1) is reached. When the compare matches, RTIFRC0/1 is incremented and RTIUC0/1 is reset to 0. If RTIFRC0/1 overflows, an interrupt is generated to the vectored interrupt manager (VIM). The overflow interrupt is not intended to generate the timebase for the operating system. See Section 13.2.2 for the timebase generation. The up counter together with the compare up counter value prescale the RTI clock. The resulting formula for the frequency of the free running counter (RTIFRC0/1) is:

$$f_{\text{RTIFRCx}} = \begin{cases} \frac{f_{\text{RTICLK}}}{\text{RTICPUCx} + 1} & \text{when RTICPUCx} \neq 0 \\ \frac{f_{\text{RTICLK}}}{2^{32} + 1} & \text{when RTICPUCx} = 0 \end{cases} \quad (23)$$

NOTE: Setting RTICPUCx equal to zero is not recommended. Doing so will hold the Up Counter at zero for two RTICLK cycles after it overflows from 0xFFFFFFFF to zero.

The counter values can be determined by reading the respective counter registers or by generating a hardware event which captures the counter value into the respective capture register. Both functions are described in the following sections.

Figure 13-2. Counter Block Diagram


13.2.1.1 Counter and Capture Read Consistency

Portions of the device internal databus are 32-bits wide. If the application wants to read the 64-bit counters or the 64-bit capture values, a certain order of 32-bit read operations needs to be followed. This is to prevent one counter incrementing in between the two separate read operations to both counters.

Reading the Counters

The free running counter (RTIFRCx) must be read first. This priority will ensure that in the cycle when the CPU reads RTIFRCx, the up counter value is stored in its counter register (RTIUCx). The second read has to access the up counter register (RTIUCx), which then holds the value which corresponds to the number of RTICLK cycles that have elapsed at the time reading the free running counter register (RTIFRCx).

NOTE: The up counters are implemented as shadow registers. Reading RTIUCx without having read RTIFRCx first will return always the same value. RTIUCx will only be updated when RTIFRCx is read.

Reading the Capture Values

The free running counter capture register (RTICAFRCx) must be read first. This priority will ensure that in the cycle when the CPU reads RTICAFRCx, the up counter value is stored in its counter register (RTICAUCx). The second read has to access the up counter register (RTICAUCx), which then holds the value captured at the time when reading the capture free running counter register (RTICAFRCx).

NOTE: The capture up counter registers are implemented as shadow registers. Reading RTICAUCx without having read RTICAFRCx first will return always the same value. RTICAUCx will only be updated when RTICAFRCx is read.

13.2.1.2 Capture Feature

Both counter blocks also provide a capture feature on external events. Two capture sources can trigger the capture event. The source triggering the block is configurable (RTICAPCTRL). The sources originate from the Vectored Interrupt Manager (VIM) and allow the generation of capture events when a peripheral modules has generated an interrupt. Any of the peripheral interrupts can be selected as the capture event in the VIM.

When an event is detected, RTIUCx and RTIFRCx are stored in the capture up counter (RTICAUCx) and capture free running counter (RTICAFRCx) registers. The read order of the captured values must be the same as the read order of the actual counters (see [Section 13.2.1.1](#)).

13.2.2 Interrupt/DMA Requests

There are four compare registers (RTICOMP_y) to generate interrupt requests to the VIM or DMA requests to the DMA controller. The interrupts can be used to generate different timebases for the operating system. Each of the compare registers can be configured to be compared to either RTIFRC0 or RTIFRC1. When the counter value matches the compare value, an interrupt is generated. To allow periodic interrupts, a certain value can be added to the compare value in RTICOMP_y automatically. This value is stored in the update compare register (RTIUDCP_y) and will be added after a compare is matched. The period of the generated interrupt/DMA request can be calculated with:

$$t_{COMPx} = t_{RTICLK} \times (RTICPUCy + 1) \times RTIUDCPy$$

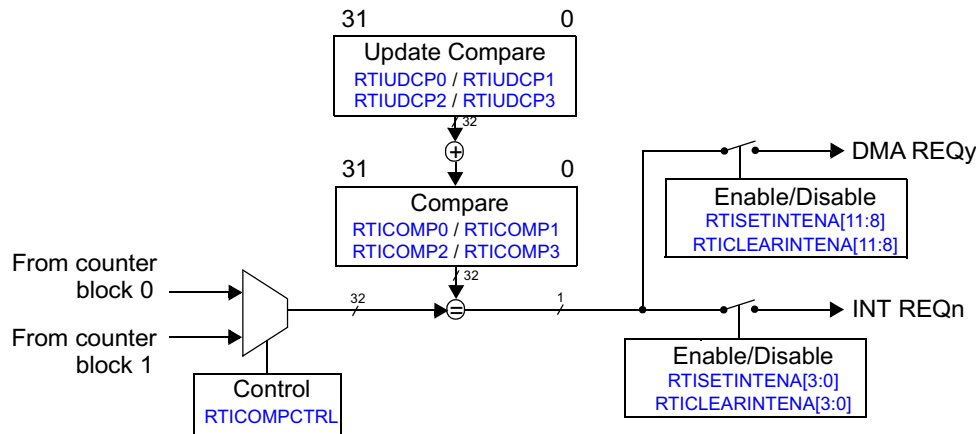
if RTICPUC_y ≠ 0,

$$t_{COMPx} = t_{RTICLK} \times (2^{32} + 1) \times RTIUDCPy$$

if RTIUDCP_y = 0,

$$t_{COMPx} = t_{RTICLK} \times (RTICPUCy + 1) \times 2^{32} \quad (24)$$

Figure 13-3. Compare Unit Block Diagram (shows only 1 of 4 blocks for simplification)



Another interrupt that can be generated is the overflow interrupt (OVLINTx) in case the RTIFRCx counter overflows.

The interrupts/DMA requests can be enabled in the RTISETINTENA register and disabled in the RTICLEARINTENA register. The RTIINTFLAG register shows the pending interrupts.

13.2.3 RTI Clocking

The counter blocks are clocked with RTICLK (for definition see [Section 2.4.2](#)).

A clock supervision for the NTUx clocking scheme is implemented to avoid missing operating system ticks.

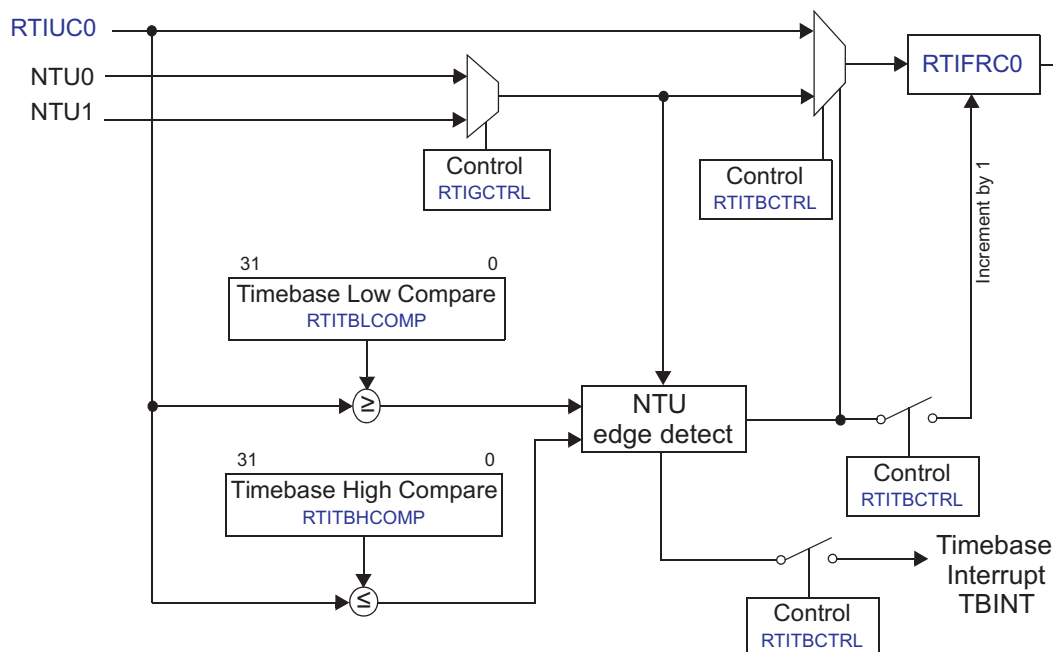
13.2.4 Synchronizing Timer Events to Network Time (NTU)

For applications which are participating on a time-triggered communication bus, it is often beneficial to synchronize the application or operating system to the network time. The RTI provides a feature to increment Free Running Counter 0 (RTIFRC0) by a periodic clock provided by the communication module. In this case two different clocks can be chosen.

The application has control over which clock (RTICLK, NTU0, NTU1) should be used for clocking RTIFRC0. If NTUx is used, a clock supervision circuit allows to monitor this clock and provides a fallback solution, should the clock be non-functional (missing). A too fast running NTUx cannot be detected.

RTIUC0 is utilized to monitor the NTUx signal. A detection window can be programmed in which a valid NTU clock pulse needs to occur. If no pulse is detected, the RTI automatically switches back to clock the Free Running Counter 0 with RTIUC0. In order to avoid a big jitter in the operating ticks, in case a switch back to RTIUC0 happens, RTICPUC0 should be set to a value so the clock frequency RTIUC0 outputs is approximately the same as the NTUx frequency.

Figure 13-4. Timebase Control



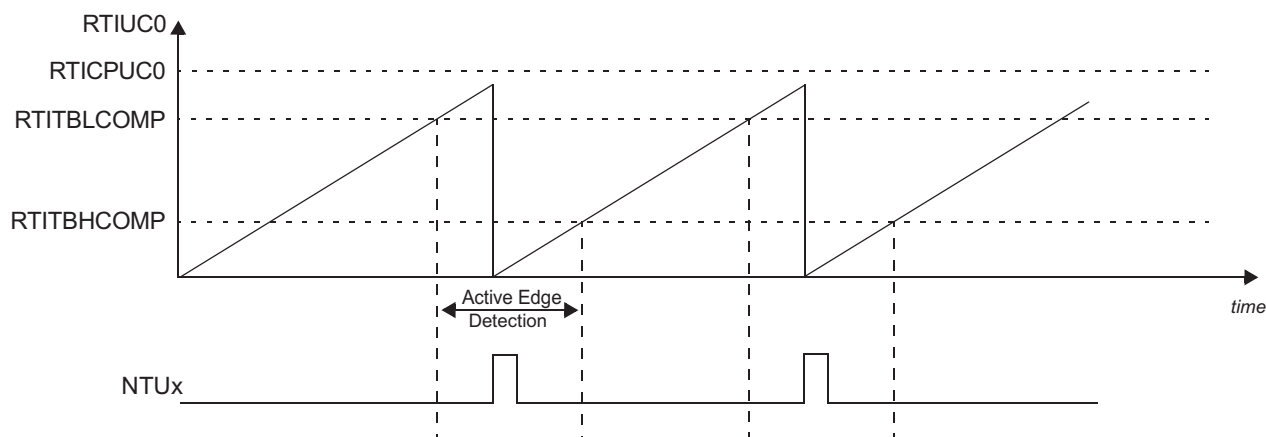
13.2.4.1 Detecting Clock Edges

To detect clock edges on the NTUx signal, the timebase low compare has to be set lower or equal than the value stored in the RTICPUC0 register and the timebase high compare has to be set higher than 0 and lower than the timebase low compare value. This effectively opens a window in which an edge of the NTUx signal is expected (see Figure 13-5). Outside this window, no edges will be detected. If no edge will occur inside the detection window, the multiplexer is switched to internal timebase. The application can select to generate a timebase interrupt (TBINT) and if the INC bit is set, also will automatically increment RTIFRC0 by one to compensate for the missed clock cycle of NTUx. If an edge occurs inside the window, RTIUC0 will be reset to synchronize the two timebases.

In order to make the edge detection work properly, the value in RTICPUC0 needs to be adapted so that RTIUC0 has a similar period as NTUx.

NOTE: To ensure the NTUx signal is properly detected, the NTUx period must be at least twice as long as the RTICLK period.

Figure 13-5. Clock Detection Scheme



13.2.4.2 Switching from Internal Source to External Source

If the application switches from an internal source to an external source, the two signals must be synchronized (see Figure 13-6). The synchronization will occur when the TBEXT bit is set. RTIUC0 will be reset and the edge detection circuit will be active for one (RTICPUC0 + RTITBHCOMP) period or until an edge is detected. If there is no pulse during this period, the source will be reset from an external clock source to an internal clock source. If an edge is detected, the windowed edge detection behavior will take place. Setting the TBEXT bit will not increment free running counter 0.

NOTE: If an external timebase is used, then the software must ensure that timebase low compare and timebase high compare are programmed to a valid state before switching TBEXT to an external source. This state is necessary to allow the timebase control circuit to operate correctly. The following condition must be met:

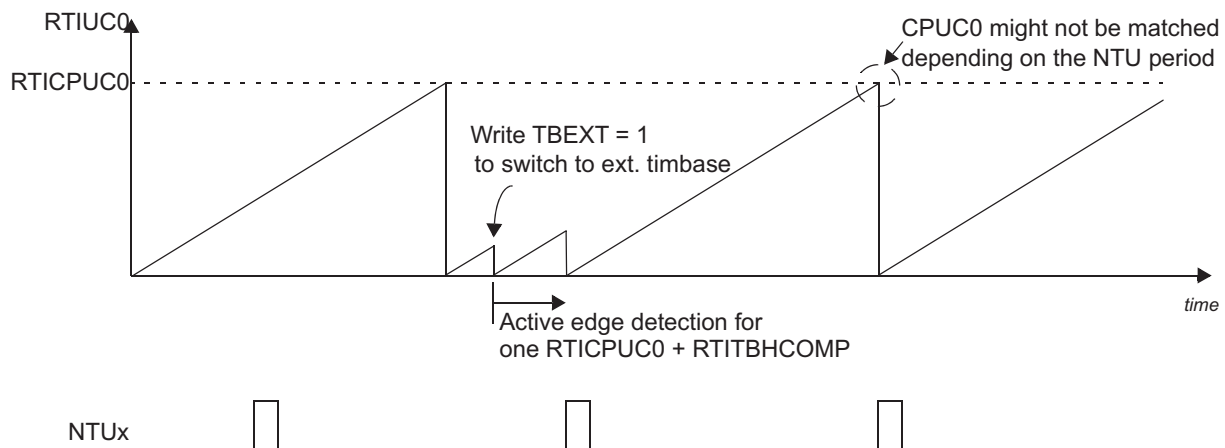
- $RTITBHCOMP < RTITBLCOMP + RTICPUC0$

RTITBHCOMP must be lower than RTICPUC0 because RTIUC0 will be reset if RTICPUC0 is reached. RTITBHCOMP will represent the number of RTICLK cycles from RTICPUC0 until the circuit switches to the internal timebase when no NTU edge is detected.

If an external timebase is used, RTIGCTRL[0] must be set to 1 (enable RTIUC0) to ensure that the timebase control circuit does not wait indefinitely for an incoming signal.

Figure 13-6 shows a timing example for the synchronization phase when the TBEXT bit is set.

Figure 13-6. Switch to NTUx

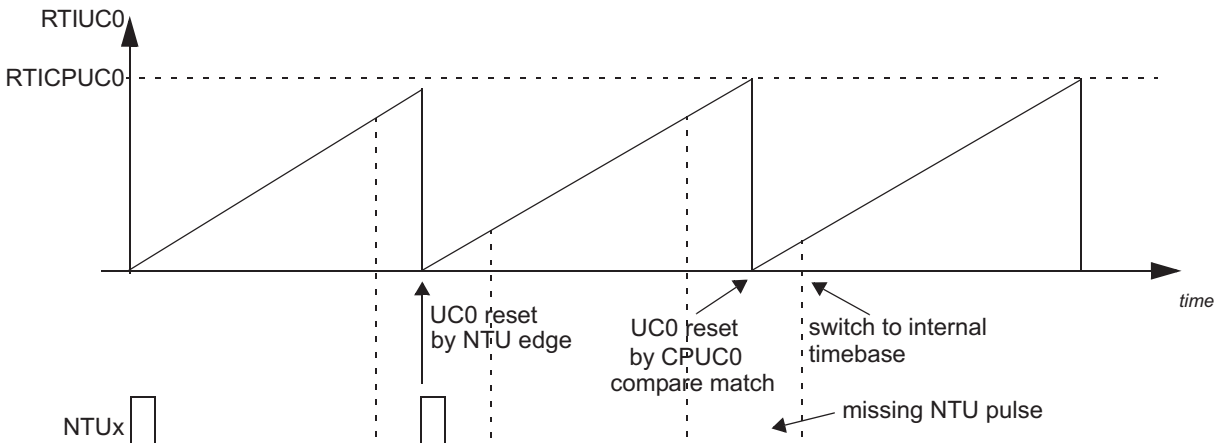


13.2.4.3 Switching from External Source to Internal Source

When the edge detection is active (TBEXT = 1) and no clock edge of NTUx is detected inside the programmed detection window, the RTI will automatically switch the timebase to RTIUC0. Figure 13-7 shows a timing example for a missing NTU signal. In the case where the INC bit is set, RTIFRC0 will automatically be incremented by one to compensate for the missed NTU pulse.

Setting TBEXT = 0 will also switch the clock source for RTIFRC0 to RTIUC0.

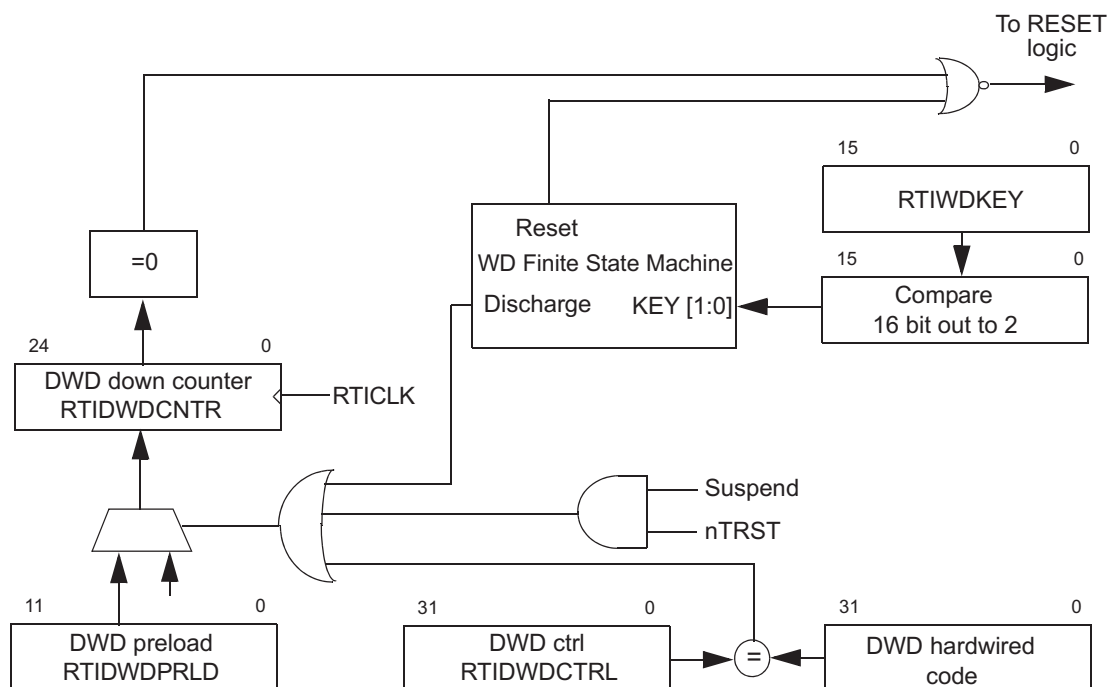
Figure 13-7. Missing NTUx Signal Example



13.2.5 Digital Watchdog (DWD)

The digital watchdog (DWD) is an optional safety diagnostic which can detect a runaway CPU and generate either a reset or NMI (non-maskable interrupt) response. It generates resets or NMIs after a programmable period, or if no correct key sequence was written to the RTIWDKEY register. Figure 13-8 illustrates the DWD.

Figure 13-8. Digital Watchdog



13.2.5.1 Digital Watchdog (DWD)

The DWD is disabled by default. If it should be used, it must be enabled by writing a 32-bit value to the RTIDWDCTRL register.

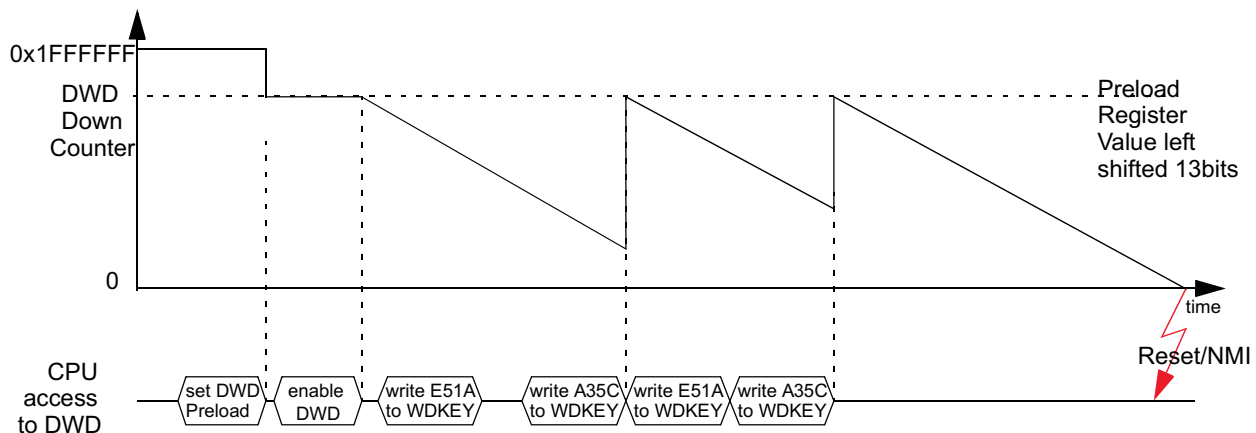
NOTE: Once the DWD is enabled, it cannot be disabled except by system reset or power on reset.

If the correct key sequence is written to the RTIWDKEY register (0xE51A followed by 0xA35C), the 25-bit DWD down counter is reloaded with the left justified 12-bit preload value stored in RTIDWDPRLD. If an incorrect value is written, a watchdog reset or NMI will occur immediately. A reset or NMI will also be generated when the DWD down counter is decremented to 0.

While the device is in suspend mode (halting debug mode), the DWD down counter keeps the value it had when entering suspend mode.

The DWD down counter will be decremented with the RTICLK frequency.

Figure 13-9. DWD Operation



The expiration time of the DWD down counter can be determined with the following equation:

$$t_{exp} = (DWDPRLD + 1) \times 2^{13} / RTICLK$$

where

$$DWDPRLD = 0 \dots 4095$$

NOTE: Care should be taken to ensure that the CPU write to the watchdog register is made allowing time for the write to propagate to the RTI.

13.2.5.2 Digital Windowed Watchdog (DWWD)

In addition to the time-out boundary configurable via the digital watchdog discussed in [Section 13.2.5.1](#), for enhanced safety metrics it is desirable to check for a watchdog "pet" within a time window rather than using a single time threshold. This is enabled by the digital windowed watchdog (DWWD) feature.

- Functional Behavior

The DWWD opens a configurable time window in which the watchdog must be serviced. Any attempt to service the watchdog outside this time window, or a failure to service the watchdog in this time window, will cause the watchdog to generate either a reset or a NMI to the CPU. This is controlled by configuring the RTIWWDRXNCTRL register. As with the DWD, the DWWD is disabled after power on reset. When the DWWD is configured to generate a non-maskable interrupt on a window violation, the watchdog counter continues to count down. The NMI handler needs to clear the watchdog violation status flag(s) and then

service the watchdog by writing the correct sequence in the watchdog key register. This service will cause the watchdog counter to get reloaded from the preload value and start counting down. If the NMI handler does not service the watchdog in time, it could count down all the way to zero and wrap around. If the NMI Handler does not service the watchdog in time, the NMI gets generated continuously, each time the counter counts to '0'.

The DWWD uses the Digital Watchdog (DWD) preload register (RTIDWDPRLD) setting to define the end-time of the window. The start-time of the window is defined by a window size configuration register(RTIWWDSIZECTRL).

The default window size is set to 100%, which corresponds to the DWD functionality of a time-out-only watchdog. The window size can be selected (through register RTIWWDSIZECTRL) from among 100%, 50%, 25%, 12.5%, 6.25% and 3.125% as shown in Figure 13-10. The window with the respective size will be opened before the end of the DWD expiration. The user has to serve the watchdog in the window. Otherwise, a reset or NMI will generate. Figure 13-11 shows an DWWD operation example (25% window).

- Configuration of DWWD

The DWWD preload value (same as DWD preload) can only be configured when the DWWD counter is disabled. The window size and watchdog reaction to a violation can be configured even after the watchdog has been enabled. Any changes to the window size and watchdog reaction configurations will only take effect after the next servicing of the DWWD. This feature can be utilized to dynamically set windows of different sizes based on task execution time, adding a program sequence element to the diagnostic which can improve fault coverage.

Figure 13-10. Digital Windowed Watchdog Timing Example

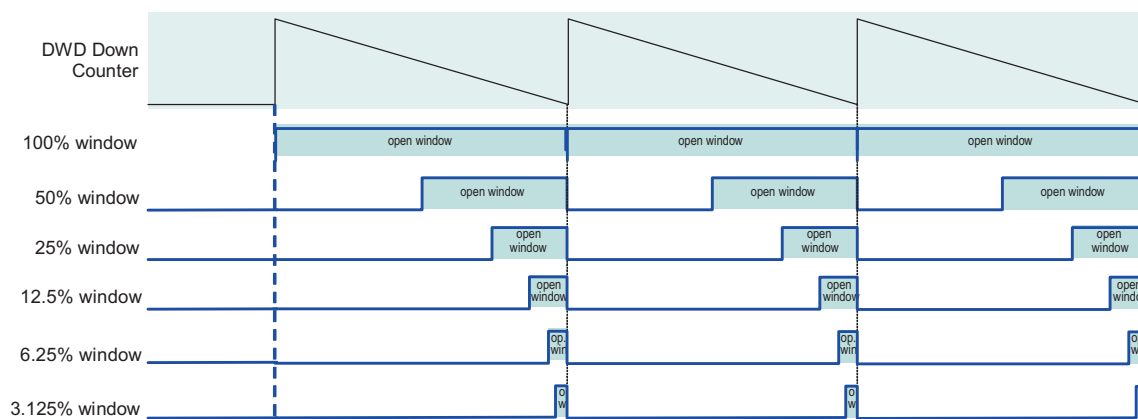
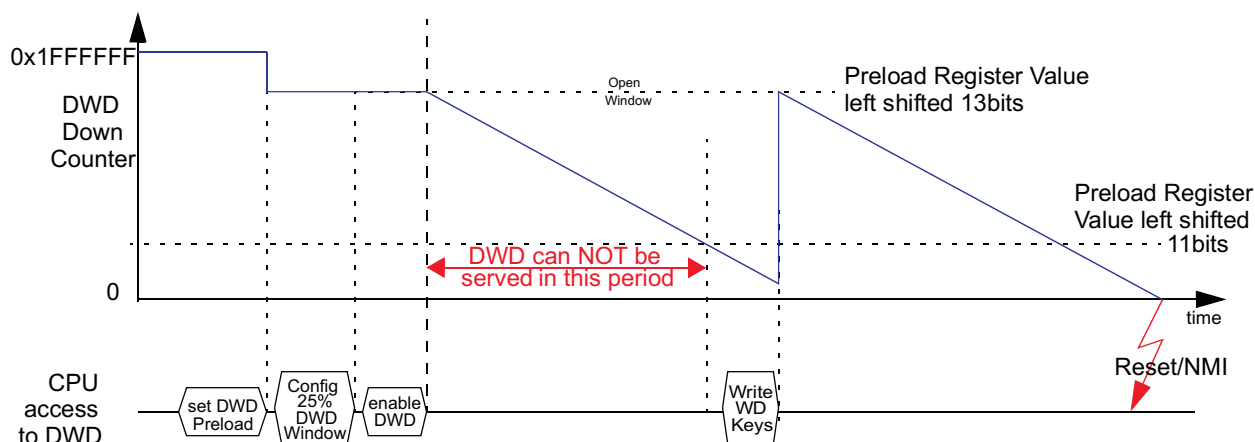


Figure 13-11. Digital Windowed Watchdog Operation Example (25% Window)



13.2.6 Low Power Modes

Low power modes allow the trade off of the current used during low power versus functionality and fast wakeup response. All low power modes have the following characteristics:

- CPU and system clocks are disabled.
- Flash banks and pump are in sleep mode.
- All peripheral modules are in low power modes and the clocks are disabled (exceptions to this may occur and would be documented in the specific device data sheet).

Flexibility in enabling and disabling clocks allows for many different low-power modes (see [Section 2.4.3](#)).

The operation of the RTI Module is guaranteed in Run, Doze and Snooze modes. In Sleep mode, all clocks will be switched off and the RTI will not work.

In Doze and Snooze modes, the RTI is active and is able to wake up the device with compare, timebase and overflow interrupts. The compare interrupts can be used to periodically wake up the device. The overflow interrupt can be used to notify the operating system that a counter overflow has occurred. Capturing events generated by the Vectored Interrupt Module (VIM) is also possible since, in both of these low power modes, the peripheral modules are able to generate interrupts that can trigger capture events. Capturing events while in Sleep mode is not supported as the clock to the RTI is not active.

When the device is put into low power mode, the peripheral which is generating the external clock NTU is no longer active, and the timebase control circuitry has to switch to an internal clocking scheme when it detects a missing clock on NTU. The timebase interrupt will wake up the device and the application software has to adapt the periodic interrupt generation to the internal clock source.

DMA transfers will be disabled, and DMA requests will not be generated after device wakeup since the DMA controller will be powered down.

NOTE: RTICK in Doze Mode

In the special case of Doze Mode with PLL off, RTICK might have a different period than with PLL enabled since RTICK will be derived from the oscillator output. It has to be ensured that the VCLK to RTICK ratio is at least 3:1.

13.2.7 Halting Debug Mode Behaviour

Once the system enters halting debug mode, the behavior of the RTI depends on the COS (continue on suspend) bit. If the bit is cleared and halting debug mode is active, all counters will stop operation. If the bit is set to one, all counters will be clocked normally and the RTI will work like in normal mode. However, if the external timebase (NTU) is used and the system is in halting debug mode, the timebase control circuit will switch to internal timebase once it detects the missing NTU signal of the suspended communication controller. This will be signaled with an TBINT interrupt so that software can resynchronize after the device exits halting debug mode.

13.3 RTI Control Registers

[Table 13-1](#) provides a summary of the registers. The registers support 8-bit, 16-bit, and 32-bit writes. The offset is relative to the associated peripheral select. See the following sections for detailed descriptions of the registers. The base address for the control registers is FFFF FC00h. The address locations not listed are reserved.

Table 13-1. RTI Registers

Offset	Acronym	Register Description	Section
00h	RTIGCTRL	RTI Global Control Register	Section 13.3.1
04h	RTITBCTRL	RTI Timebase Control Register	Section 13.3.2
08h	RTICAPCTRL	RTI Capture Control Register	Section 13.3.3
0Ch	RTICOMPCTRL	RTI Compare Control Register	Section 13.3.4
10h	RTIFRC0	RTI Free Running Counter 0 Register	Section 13.3.5
14h	RTIUC0	RTI Up Counter 0 Register	Section 13.3.6
18h	RTICPUC0	RTI Compare Up Counter 0 Register	Section 13.3.7
20h	RTICAFRC0	RTI Capture Free Running Counter 0 Register	Section 13.3.8
24h	RTICAUC0	RTI Capture Up Counter 0 Register	Section 13.3.9
30h	RTIFRC1	RTI Free Running Counter 1 Register	Section 13.3.10
34h	RTIUC1	RTI Up Counter 1 Register	Section 13.3.11
38h	RTICPUC1	RTI Compare Up Counter 1 Register	Section 13.3.12
40h	RTICAFRC1	RTI Capture Free Running Counter 1 Register	Section 13.3.13
44h	RTICAUC1	RTI Capture Up Counter 1 Register	Section 13.3.14
50h	RTICOMP0	RTI Compare 0 Register	Section 13.3.15
54h	RTIUDCP0	RTI Update Compare 0 Register	Section 13.3.16
58h	RTICOMP1	RTI Compare 1 Register	Section 13.3.17
5Ch	RTIUDCP1	RTI Update Compare 1 Register	Section 13.3.18
60h	RTICOMP2	RTI Compare 2 Register	Section 13.3.19
64h	RTIUDCP2	RTI Update Compare 2 Register	Section 13.3.20
68h	RTICOMP3	RTI Compare 3 Register	Section 13.3.21
6Ch	RTIUDCP3	RTI Update Compare 3 Register	Section 13.3.22
70h	RTITBLCOMP	RTI Timebase Low Compare Register	Section 13.3.23
74h	RTITBHCMP	RTI Timebase High Compare Register	Section 13.3.24
80h	RTISETINTENA	RTI Set Interrupt Enable Register	Section 13.3.25
84h	RTICLEARINTENA	RTI Clear Interrupt Enable Register	Section 13.3.26
88h	RTIINTFLAG	RTI Interrupt Flag Register	Section 13.3.27
90h	RTIDWDCTRL	Digital Watchdog Control Register	Section 13.3.28
94h	RTIDWDPRLD	Digital Watchdog Preload Register	Section 13.3.29
98h	RTIWDSTATUS	Watchdog Status Register	Section 13.3.30
9Ch	RTIWDKEY	RTI Watchdog Key Register	Section 13.3.31
A0h	RTIDWDCNTR	RTI Digital Watchdog Down Counter Register	Section 13.3.32
A4h	RTIWWDRXNCTRL	Digital Windowed Watchdog Reaction Control Register	Section 13.3.33
A8h	RTIWWDSIZECTRL	Digital Windowed Watchdog Window Size Control Register	Section 13.3.34
ACH	RTIINTCLRENABLE	RTI Compare Interrupt Clear Enable Register	Section 13.3.35
B0h	RTICOMP0CLR	RTI Compare 0 Clear Register	Section 13.3.36
B4h	RTICOMP1CLR	RTI Compare 1 Clear Register	Section 13.3.37
B8h	RTICOMP2CLR	RTI Compare 2 Clear Register	Section 13.3.38
BCh	RTICOMP3CLR	RTI Compare 3 Clear Register	Section 13.3.39

NOTE: Writes to Reserved registers may clear the pending RTI interrupt.

13.3.1 RTI Global Control Register (RTIGCTRL)

The global control register starts/stops the counters and selects the signal compared with the timebase control circuit. This register is shown in [Figure 13-12](#) and described in [Table 13-2](#).

Figure 13-12. RTI Global Control Register (RTIGCTRL) [offset = 00]

31		20	19		16
Reserved				NTUSEL	
R-0				R/WP-0	
15	14		2	1	0
COS	Reserved			CNT1EN	CNT0EN
R/WP-0		R-0		R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-2. RTI Global Control Register (RTIGCTRL) Field Descriptions

Bit	Field	Value	Description
31-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	NTUSEL	0h 5h Ah Fh All other values	Select NTU signal. These bits determine which NTU input signal is used as external timebase NTU0 NTU1 NTU2 NTU3 Tied to 0
15	COS	0 1	Continue on suspend. This bit determines if both counters are stopped when the device goes into halting debug mode or if they continue counting. Counters are stopped while in halting debug mode. Counters are running while in halting debug mode.
14-2	Reserved	0	Reads return 0. Writes have no effect.
1	CNT1EN	0 1	Counter 1 enable. This bit starts and stops counter block 1 (RTIUC1 and RTIFRC1). Counter block 1 is stopped. Counter block 1 is running.
0	CNT0EN	0 1	Counter 0 enable. This bit starts and stops counter block 0 (RTIUC0 and RTIFRC0). Counter block 0 is stopped. Counter block 0 is running.

NOTE: If the application uses the timebase circuit for synchronization between the communications controller and the operating system and the device enters halting debug mode, the synchronization may be lost depending on the COS setting in the RTI module and the halting debug mode behavior of the communications controller.

13.3.2 RTI Timebase Control Register (RTITBCTRL)

The timebase control register selects if the free running counter 0 is incremented by RTICLK or NTU. This register is shown in [Figure 13-13](#) and described in [Table 13-3](#).

Figure 13-13. RTI Timebase Control Register (RTITBCTRL) [offset = 04h]

31																													8
Reserved																													
R-0																													
7															2	1	0												
Reserved														INC		TBEXT													
R-0														R/WP-0		R/WP-0													

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

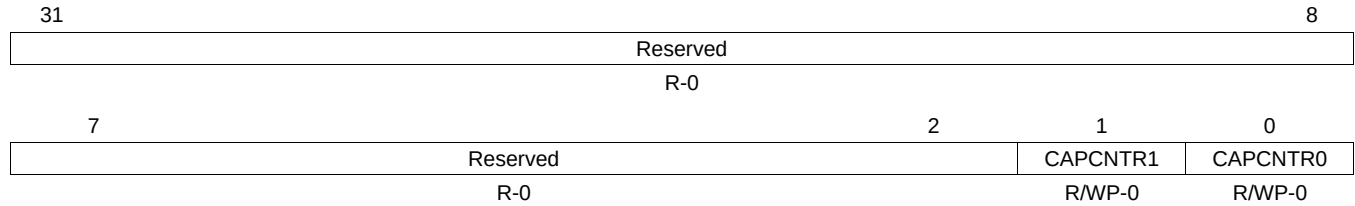
Table 13-3. RTI Timebase Control Register (RTITBCTRL) Field Descriptions

Bit	Field	Value	Description
31-2	Reserved	0	Reads return 0. Writes have no effect.
1	INC	0 1	Increment free running counter 0. This bit determines whether the free running counter 0 (RTIFRC0) is automatically incremented if a failing clock on the NTU signal is detected. RTIFRC0 will not be incremented on a failing external clock. RTIFRC0 will be incremented on a failing external clock.
0	TBEXT	0 1	Timebase external. This bit selects whether the free running counter 0 (RTIFRC0) is clocked by the internal up counter 0 (RTIUC0) or from the external signal NTU. Setting the TBEXT bit from 0 to 1 will not increment RTIFRC0, since RTIUC0 is reset. When the timebase supervisor circuit detects a missing clock edge, then the TBEXT bit is reset. Only the software can select whether the external signal should be used. RTIUC0 clocks RTIFRC0. NTU clocks RTIFRC0.

13.3.3 RTI Capture Control Register (RTICAPCTRL)

The capture control register controls the capture source for the counters. This register is shown in [Figure 13-14](#) and described in [Table 13-4](#).

Figure 13-14. RTI Capture Control Register (RTICAPCTRL) [offset = 08h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-4. RTI Capture Control Register (RTICAPCTRL) Field Descriptions

Bit	Field	Value	Description
31-2	Reserved	0	Reads return 0. Writes have no effect.
1	CAPCNTR1	0	Capture counter 1. This bit determines which external interrupt source triggers a capture event of RTIUC1 and RTIFRC1.
		0	Capture of RTIUC1/ RTIFRC1 is triggered by capture event source 0.
		1	Capture of RTIUC1/ RTIFRC1 is triggered by capture event source 1.
0	CAPCNTR0	0	Capture counter 0. This bit determines which external interrupt source triggers a capture event of RTIUC0 and RTIFRC0.
		0	Capture of RTIUC0/ RTIFRC0 is triggered by capture event source 0.
		1	Capture of RTIUC0/ RTIFRC0 is triggered by capture event source 1.

13.3.4 RTI Compare Control Register (RTICOMPCTRL)

The compare control register controls the source for the compare registers. This register is shown in [Figure 13-15](#) and described in [Table 13-5](#).

Figure 13-15. RTI Compare Control Register (RTICOMPCTRL) [offset = 0Ch]

31	Reserved																16
R-0																	
15	Reserved				13	12	11	Reserved				9	Reserved				8
R-0				COMPSEL3				R-0				COMPSEL2				R/WP-0	
7	Reserved				5	4	3	Reserved				1	Reserved				0
R-0				COMPSEL1				R-0				COMPSEL0				R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-5. RTI Compare Control Register (RTICOMPCTRL) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12	COMPSEL3	0	Compare select 3. This bit determines the counter with which the compare value held in compare register 3 (RTICOMP3) is compared. Value will be compared with RTIFRC0.
		1	Value will be compared with RTIFRC1.
11-9	Reserved	0	Reads return 0. Writes have no effect.
8	COMPSEL2	0	Compare select 2. This bit determines the counter with which the compare value held in compare register 2 (RTICOMP2) is compared. Value will be compared with RTIFRC0.
		1	Value will be compared with RTIFRC1.
7-5	Reserved	0	Reads return 0. Writes have no effect.
4	COMPSEL1	0	Compare select 1. This bit determines the counter with which the compare value held in compare register 1 (RTICOMP1) is compared. Value will be compared with RTIFRC0.
		1	Value will be compared with RTIFRC1.
3-1	Reserved	0	Reads return 0. Writes have no effect.
0	COMPSEL0	0	Compare select 0. This bit determines the counter with which the compare value held in compare register 0 (RTICOMP0) is compared. Value will be compared with RTIFRC0.
		1	Value will be compared with RTIFRC1.

13.3.5 RTI Free Running Counter 0 Register (RTIFRC0)

The free running counter 0 register holds the current value of free running counter 0. This register is shown in [Figure 13-16](#) and described in [Table 13-6](#).

Figure 13-16. RTI Free Running Counter 0 Register (RTIFRC0) [offset = 10h]

31	16
FRC0	
R/WP-0	
15	0
FRC0	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-6. RTI Free Running Counter 0 Register (RTIFRC0) Field Descriptions

Bit	Field	Value	Description
31-0	FRC0	0-FFFF FFFFh	Free running counter 0. This registers holds the current value of the free running counter 0. A read of this counter returns the current value of the counter. The counter can be preset by writing (in privileged mode only) to this register. The counter increments then from this written value upwards. Note: If counters must be preset, they must be disabled in the RTIGCTRL register to ensure consistency between RTIUC0 and RTIFRC0.

13.3.6 RTI Up Counter 0 Register (RTIUC0)

The up counter 0 register holds the current value of prescale counter. This register is shown in [Figure 13-17](#) and described in [Table 13-7](#).

Figure 13-17. RTI Up Counter 0 Register (RTIUC0) [offset = 14h]

31	16
UC0	
R/WP-0	
15	0
UC0	
R/WP-0	

LLEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-7. RTI Up Counter 0 Register (RTIUC0) Field Descriptions

Bit	Field	Value	Description
31-0	UC0	0-FFFF FFFFh	Up counter 0. This register holds the current value of the up counter 0 and prescales the RTI clock. It will be only updated by a previous read of free running counter 0 (RTIFRC0). This method of updating effectively gives a 64-bit read of both counters, without having the problem of a counter being updated between two consecutive reads on up counter 0 (RTIUC0) and free running counter 0 (RTIFRC0). A read of this counter returns the value of the counter at the time RTIFRC0 was read. A write to this counter presets it with a value. The counter then increments from this written value upwards. Note: If counters must be preset, they must be disabled in the RTIGCTRL register to ensure consistency between RTIUC0 and RTIFRC0. Note: If the preset value is bigger than the compare value stored in register RTICPUC0, then it can take a long time until a compare matches, since RTIUC0 has to count up until it overflows.

13.3.7 RTI Compare Up Counter 0 Register (RTICPUC0)

The compare up counter 0 register holds the value to be compared with prescale counter 0 (RTIUC0). This register is shown in [Figure 13-18](#) and described in [Table 13-8](#).

Figure 13-18. RTI Compare Up Counter 0 Register (RTICPUC0) [offset = 18h]

31	16
CPUC0	
R/WP-0	
15	0
CPUC0	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-8. RTI Compare Up Counter 0 Register (RTICPUC0) Field Descriptions

Bit	Field	Value	Description
31-0	CPUC0	0-FFFF FFFFh	Compare up counter 0. This register holds the value that is compared with the up counter 0. When the compare shows a match, the free running counter 0 (RTIFRC0) is incremented. RTIUC0 is set to 0 when the counter value matches the RTICPUC0 value. The value set in this register prescales the RTI clock. If CPUC0 = 0, then $f_{\text{RC0}} = \text{RTICLK}/(2^{32}+1)$ (Setting CPUC0 equal to 0 is not recommended. Doing so will hold the Up Counter at 0 for 2 RTICLK cycles after it overflows from FFFF FFFFh to 0.) If CPUC0 $\neq$ 0, then $f_{\text{RC0}} = \text{RTICLK}/(\text{RTICPUC0}+1)$ A read of this register returns the current compare value. A write to this register: - If TBEXT = 0, the compare value is updated. - If TBEXT = 1, the compare value is unchanged.

13.3.8 RTI Capture Free Running Counter 0 Register (RTICAFRC0)

The capture free running counter 0 register holds the free running counter 0 on external events. This register is shown in [Figure 13-19](#) and described in [Table 13-9](#).

Figure 13-19. RTI Capture Free Running Counter 0 Register (RTICAFRC0) [offset = 20h]

31	16
CAFRC0	
R-0	
15	0
CAFRC0	
R-0	

LEGEND: R = Read only; -n = value after reset

Table 13-9. RTI Capture Free Running Counter 0 Register (RTICAFRC0) Field Descriptions

Bit	Field	Value	Description
31-0	CAFRC0	0-FFFF FFFFh	Capture free running counter 0. This register captures the current value of the free running counter 0 (RTIFRC0) when an event occurs, controlled by the external capture control block. A read of this register returns the value of RTIFRC0 on a capture event.

13.3.9 RTI Capture Up Counter 0 Register (RTICAUC0)

The capture up counter 0 register holds the current value of prescale counter 0 on external events. This register is shown in [Figure 13-20](#) and described in [Table 13-10](#).

Figure 13-20. RTI Capture Up Counter 0 Register (RTICAUC0) [offset = 24h]

31	16
CAUC0	
R-0	
15	0
CAUC0	
R-0	

LEGEND: R = Read only; -n = value after reset

Table 13-10. RTI Capture Up Counter 0 Register (RTICAUC0) Field Descriptions

Bit	Field	Value	Description
31-0	CAUC0	0-FFFF FFFFh	Capture up counter 0. This register captures the current value of the up counter 0 (RTIUC0) when an event occurs, controlled by the external capture control block. Note: The read sequence must be the same as with RTIUC0 and RTIFRC0. Therefore, the RTICAFRC0 register must be read before the RTICAUC0 register is read. This sequence ensures that the value of the RTICAUC0 register is the corresponding value to the RTICAFRC0 register, even if another capture event happens in between the two reads. A read of this register returns the value of RTIUC0 on a capture event.

13.3.10 RTI Free Running Counter 1 Register (RTIFRC1)

The free running counter 1 register holds the current value of the free running counter 1. This register is shown in [Figure 13-21](#) and described in [Table 13-11](#).

Figure 13-21. RTI Free Running Counter 1 Register (RTIFRC1) [offset = 30h]

31	16
FRC1	
R/WP-0	
15	0
FRC1	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

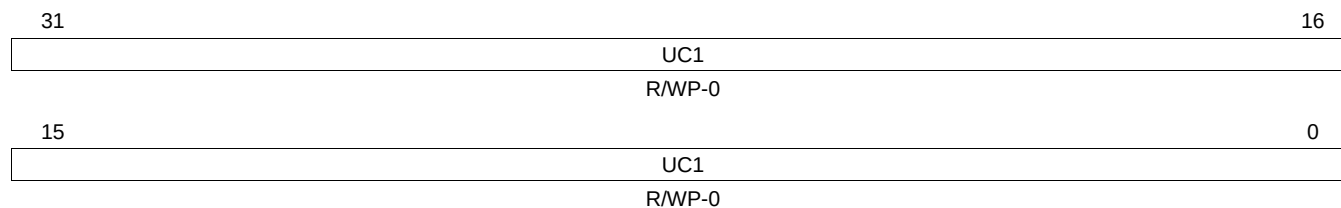
Table 13-11. RTI Free Running Counter 1 Register (RTIFRC1) Field Descriptions

Bit	Field	Value	Description
31-0	FRC1	0-FFFF FFFFh	Free running counter 1. This register holds the current value of the free running counter 1 and will be updated continuously. A read of this register returns the current value of the counter. A write to this register presets the counter. The counter increments then from this written value upwards. Note: If counters must be preset, they must be disabled in the RTIGCTRL register to ensure consistency between RTIUC1 and RTIFRC1.

13.3.11 RTI Up Counter 1 Register (RTIUC1)

The up counter 1 register holds the current value of the prescale counter 1. This register is shown in [Figure 13-22](#) and described in [Table 13-12](#).

Figure 13-22. RTI Up Counter 1 Register (RTIUC1) [offset = 34h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

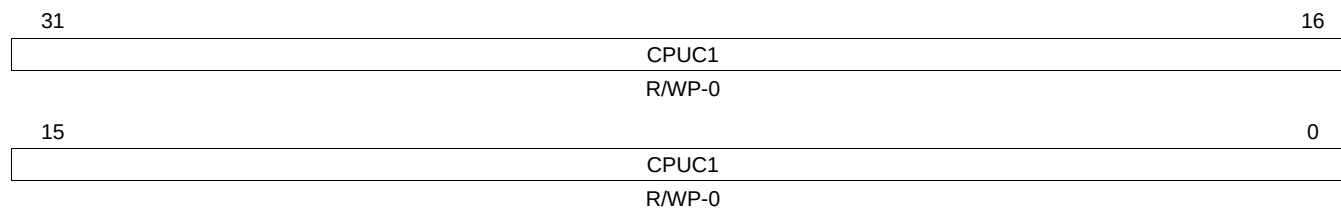
Table 13-12. RTI Up Counter 1 Register (RTIUC1) Field Descriptions

Bit	Field	Value	Description
31-0	UC1	0-FFFF FFFFh	Up counter 1. This register holds the current value of the up counter 1 and prescales the RTI clock. It will be only updated by a previous read of free running counter 1 (RTIFRC1). This method of updating effectively gives a 64-bit read of both counters, without having the problem of a counter being updated between two consecutive reads on RTIUC1 and RTIFRC1. A read of this register will return the value of the counter when the RTIFRC1 was read. A write to this register presets the counter. The counter then increments from this written value upwards. Note: If counters must be preset, they must be disabled in the RTIGCTRL register to ensure consistency between RTIUC1 and RTIFRC1. Note: If the preset value is bigger than the compare value stored in register RTICPUC1, then it can take a long time until a compare matches, since RTIUC1 has to count up until it overflows.

13.3.12 RTI Compare Up Counter 1 Register (RTICPUC1)

The compare up counter 1 register holds the value compared with prescale counter 1. This register is shown in [Figure 13-23](#) and described in [Table 13-13](#).

Figure 13-23. RTI Compare Up Counter 1 Register (RTICPUC1) [offset = 38h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

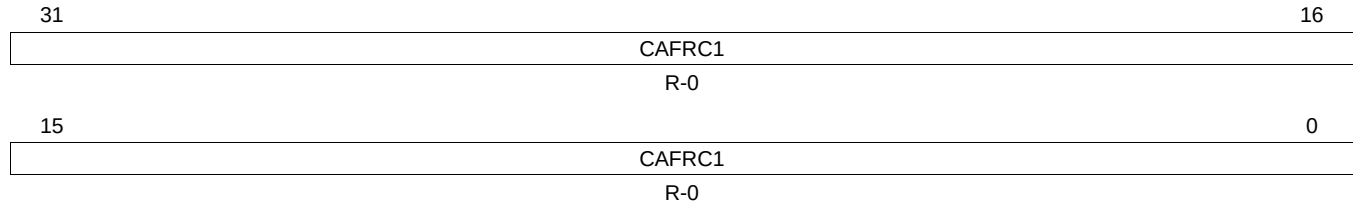
Table 13-13. RTI Compare Up Counter 1 Register (RTICPUC1) Field Descriptions

Bit	Field	Value	Description
31-0	CPUC1	0-FFFF FFFFh	Compare up counter 1. This register holds the compare value, which is compared with the up counter 1. When the compare matches, the free running counter 1 (RTIFRC1) is incremented. The up counter is cleared to 0 when the counter value matches the CPUC1 value. The value set in this prescales the RTI clock according to the following formula: If CPUC1 = 0, then $f_{FRC1} = RTICLK / (2^{32} + 1)$ (Setting CPUC1 equal to 0 is not recommended. Doing so will hold the Up Counter at 0 for 2 RTICLK cycles after it overflows from FFFF FFFFh to 0.) If CPUC1 $\neq$ 0, then $f_{FRC1} = RTICLK / (RTICPUC1 + 1)$ A read of this register returns the current compare value. A write to this register updates the compare value.

13.3.13 RTI Capture Free Running Counter 1 Register (RTICAFRC1)

The capture free running counter 1 register holds the current value of free running counter 1 on external events. This register is shown in [Figure 13-24](#) and described in [Table 13-14](#).

Figure 13-24. RTI Capture Free Running Counter 1 Register (RTICAFRC1) [offset = 40h]



LEGEND: R = Read only; -n = value after reset

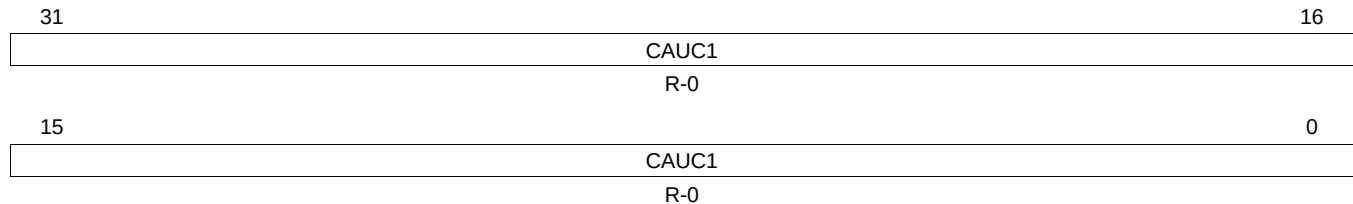
Table 13-14. RTI Capture Free Running Counter 1 Register (RTICAFRC1) Field Descriptions

Bit	Field	Value	Description
31-0	CAFRC1	0-FFFF FFFFh	Capture free running counter 1. This register captures the current value of the free running counter 1 (RTIFRC1) when an event occurs, controlled by the external capture control block. A read of this register returns the value of RTIFRC1 on a capture event.

13.3.14 RTI Capture Up Counter 1 Register (RTICAUC1)

The capture up counter 1 register holds the current value of prescale counter 1 on external events. This register is shown in [Figure 13-25](#) and described in [Table 13-15](#).

Figure 13-25. RTI Capture Up Counter 1 Register (RTICAUC1) [offset = 44h]



LEGEND: R = Read only; -n = value after reset

Table 13-15. RTI Capture Up Counter 1 Register (RTICAUC1) Field Descriptions

Bit	Field	Value	Description
31-0	CAUC1	0-FFFF FFFFh	Capture up counter 1. This register captures the current value of the up counter 1 (RTIUC1) when an event occurs, controlled by the external capture control block. Note: The RTICAFRC1 register must be read before the RTICAUC1 register is read. This sequence ensures that the value of the RTICAUC1 register is the corresponding value to the RTICAFRC1 register, even if another capture event happens in between the two reads. A read of this register returns the value of RTIUC1 on a capture event.

13.3.15 RTI Compare 0 Register (RTICOMP0)

The compare 0 register holds the value to be compared with the counters. This register is shown in [Figure 13-26](#) and described in [Table 13-16](#).

Figure 13-26. RTI Compare 0 Register (RTICOMP0) [offset = 50h]

31	16
COMP0	
R/WP-0	
15	0
COMP0	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-16. RTI Compare 0 Register (RTICOMP0) Field Descriptions

Bit	Field	Value	Description
31-0	COMP0	0-FFFF FFFFh	Compare 0. This registers holds a value that is compared with the counter selected in the compare control logic. If RTIFRC0 or RTIFRC1, depending on the counter selected, matches the compare value, an interrupt is flagged. With this register it is also possible to initiate a DMA request. A read of this register will return the current compare value. A write to this register (in privileged mode only) will update the compare register with a new compare value.

13.3.16 RTI Update Compare 0 Register (RTIUDCP0)

The update compare 0 register holds the value to be added to the compare register 0 value on a compare match. This register is shown in [Figure 13-27](#) and described in [Table 13-17](#).

Figure 13-27. RTI Update Compare 0 Register (RTIUDCP0) [offset = 54h]

31	16
UDCP0	
R/WP-0	
15	0
UDCP0	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-17. RTI Update Compare 0 Register (RTIUDCP0) Field Descriptions

Bit	Field	Value	Description
31-0	UDCP0	0-FFFF FFFFh	Update compare 0. This register holds a value that is added to the value in the compare 0 (RTICOMP0) register each time a compare matches. This function allows periodic interrupts to be generated without software intervention. A read of this register will return the value to be added to the RTICOMP0 register on the next compare match. A write to this register will provide a new update value.

13.3.17 RTI Compare 1 Register (RTICOMP1)

The compare 1 register holds the value to be compared to the counters. This register is shown in [Figure 13-28](#) and described in [Table 13-18](#).

Figure 13-28. RTI Compare 1 Register (RTICOMP1) [offset = 58h]

31	16
COMP1	
R/WP-0	
15	0
COMP1	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-18. RTI Compare 1 Register (RTICOMP1) Field Descriptions

Bit	Field	Value	Description
31-0	COMP1	0-FFFF FFFFh	Compare 1. This register holds a value that is compared with the counter selected in the compare control logic. If RTIFRC0 or RTIFRC1, depending on the counter selected, matches this compare value, an interrupt is flagged. With this register, it is possible to initiate a DMA request. A read of this register will return the current compare value. A write to this register will update the compare register with a new compare value.

13.3.18 RTI Update Compare 1 Register (RTIUDCP1)

The update compare 1 register holds the value to be added to the compare register 1 value on a compare match. This register is shown in [Figure 13-29](#) and described in [Table 13-19](#).

Figure 13-29. RTI Update Compare 1 Register (RTIUDCP1) [offset = 5Ch]

31	16
UDCP1	
R/WP-0	
15	0
UDCP1	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-19. RTI Update Compare 1 Register (RTIUDCP1) Field Descriptions

Bit	Field	Value	Description
31-0	UDCP1	0-FFFF FFFFh	Update compare 1. This register holds a value that is added to the value in the RTICOMP1 register each time a compare matches. This process allows periodic interrupts to be generated without software intervention. A read of this register will return the value to be added to the RTICOMP1 register on the next compare match. A write to this register will provide a new update value.

13.3.19 RTI Compare 2 Register (RTICOMP2)

The compare 2 register holds the value to be compared to the counters. This register is shown in [Figure 13-30](#) and described in [Table 13-20](#).

Figure 13-30. RTI Compare 2 Register (RTICOMP2) [offset = 60h]

31	16
COMP2	
R/WP-0	
15	0
COMP2	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-20. RTI Compare 2 Register (RTICOMP2) Field Descriptions

Bit	Field	Value	Description
31-0	COMP2	0-FFFF FFFFh	Compare 2. This register holds a value that is compared with the counter selected in the compare control logic. If RTIFRC0 or RTIFRC1, depending on the counter selected, matches this compare value, an interrupt is flagged. With this register, it is possible to initiate a DMA request. A read of this register will return the current compare value. A write to this register (in privileged mode only) will provide a new compare value.

13.3.20 RTI Update Compare 2 Register (RTIUDCP2)

The update compare 2 register holds the value to be added to the compare register 2 value on a compare match. This register is shown in [Figure 13-31](#) and described in [Table 13-21](#).

Figure 13-31. RTI Update Compare 2 Register (RTIUDCP2) [offset = 64h]

31	16
UDCP2	
R/WP-0	
15	0
UDCP2	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-21. RTI Update Compare 2 Register (RTIUDCP2) Field Descriptions

Bit	Field	Value	Description
31-0	UDCP2	0-FFFF FFFFh	Update compare 2. This register holds a value that is added to the value in the RTICOMP2 register each time a compare matches. This process makes it possible to generate periodic interrupts without software intervention. A read of this register will return the value to be added to the RTICOMP2 register on the next compare match. A write to this register will provide a new update value.

13.3.21 RTI Compare 3 Register (RTICOMP3)

The compare 3 register holds the value to be compared to the counters. This register is shown in [Figure 13-32](#) and described in [Table 13-22](#).

Figure 13-32. RTI Compare 3 Register (RTICOMP3) [offset = 68h]

31	16
COMP3	
R/WP-0	
15	0
COMP3	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-22. RTI Compare 3 Register (RTICOMP3) Field Descriptions

Bit	Field	Value	Description
31-0	COMP3	0-FFFF FFFFh	Compare 3. This register holds a value that is compared with the counter selected in the compare control logic. If RTIFRC0 or RTIFRC1, depending on the counter selected, matches this compare value, an interrupt is flagged. With this register, it is possible to initiate a DMA request. A read of this register will return the current compare value. A write to this register will provide a new compare value.

13.3.22 RTI Update Compare 3 Register (RTIUDCP3)

The update compare 3 register holds the value to be added to the compare register 3 value on a compare match. This register is shown in [Figure 13-33](#) and described in [Table 13-23](#).

Figure 13-33. RTI Update Compare 3 Register (RTIUDCP3) [offset = 6Ch]

31	16
UDCP3	
R/WP-0	
15	0
UDCP3	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

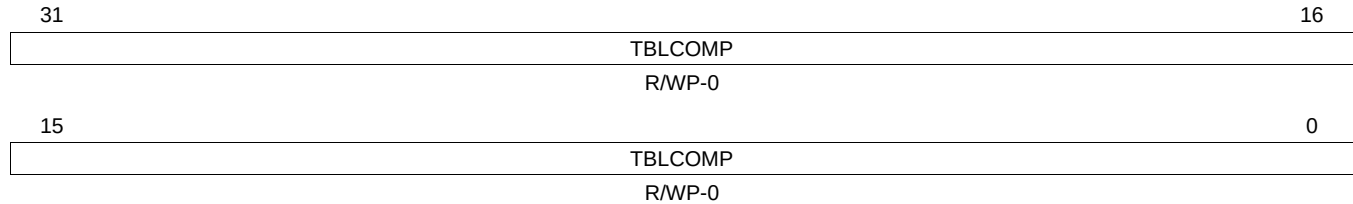
Table 13-23. RTI Update Compare 3 Register (RTIUDCP3) Field Descriptions

Bit	Field	Value	Description
31-0	UDCP3	0-FFFF FFFFh	Update compare 3. This register holds a value that is added to the value in the RTICOMP3 register each time a compare matches. This process makes it possible to generate periodic interrupts without software intervention. A read of this register will return the value to be added to the RTICOMP3 register on the next compare match. A write to this register will provide a new update value.

13.3.23 RTI Timebase Low Compare Register (RTITBLCOMP)

The timebase low compare register holds the value to activate the edge detection circuit. This register is shown in [Figure 13-34](#) and described in [Table 13-24](#).

Figure 13-34. RTI Timebase Low Compare Register (RTITBLCOMP) [offset = 70h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

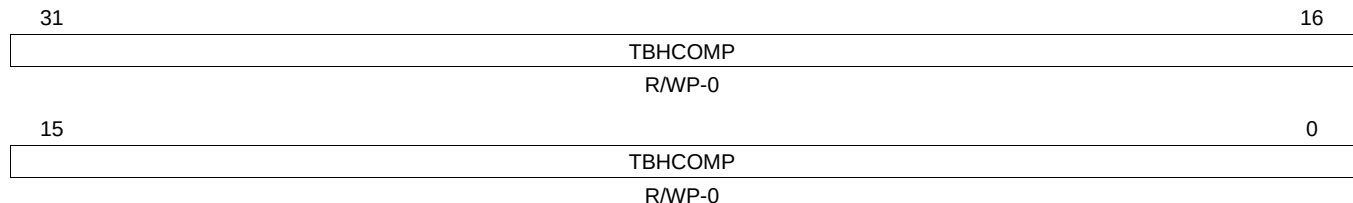
Table 13-24. RTI Timebase Low Compare Register (RTITBLCOMP) Field Descriptions

Bit	Field	Value	Description
31-0	TBLCOMP	0-FFFF FFFFh	Timebase low compare value. This value determines when the edge detection circuit starts monitoring the NTU signal. It will be compared with RTIUC0. A read of this register will return the current compare value. A write to this register has the following effects: If TBEXT = 0: The compare value is updated. If TBEXT = 1: The compare value is not changed.

13.3.24 RTI Timebase High Compare Register (RTITBHCMP)

The timebase high compare register holds the value to deactivate the edge detection circuit. This register is shown in [Figure 13-35](#) and described in [Table 13-25](#).

Figure 13-35. RTI Timebase High Compare Register (RTITBHCMP) [offset = 74h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-25. RTI Timebase High Compare Register (RTITBHCMP) Field Descriptions

Bit	Field	Value	Description
31-0	TBHCMP	0-FFFF FFFFh	Timebase high compare value. This value determines when the edge detection circuit will stop monitoring the NTU signal. It will be compared with RTIUC0. RTITBHCMP must be less than RTICPUC0 because RTIUC0 will be reset when RTICPUC0 is reached. Example: The NTU edge detection circuit should be active ± 10 RTICLK cycles around RTICPUC0. - RTICPUC0 = 0050h - RTITBLCOMP = 0046h - RTITBHCMP = 0009h A read of this register will return the current compare value. A write to this register has the following effects: If TBEXT = 0: The compare value is updated. If TBEXT = 1: The compare value is not changed.

13.3.25 RTI Set Interrupt Enable Register (RTISETINTENA)

This register prevents the necessity of a read-modify-write operation if a particular interrupt should be enabled. This register is shown in [Figure 13-36](#) and described in [Table 13-26](#).

Figure 13-36. RTI Set Interrupt Control Register (RTISETINTENA) [offset = 80h]

31																	24
Reserved																	
R-0																	
23											19	18	17	16			
Reserved											SETOVL1INT	SETOVL0INT	SETTBINT				
R-0											R/WP-0	R/WP-0	R/WP-0				
15											12	11	10	9	8		
Reserved											SETDMA3	SETDMA2	SETDMA1	SETDMA0			
R-0											R/WP-0	R/WP-0	R/WP-0	R/WP-0			
7											4	3	2	1	0		
Reserved											SETINT3	SETINT2	SETINT1	SETINT0			
R-0											R/WP-0	R/WP-0	R/WP-0	R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-26. RTI Set Interrupt Control Register (RTISETINTENA) Field Descriptions

Bit	Field	Value	Description
31-19	Reserved	0	Reads return 0. Writes have no effect.
18	SETOVL1INT	0	Set free running counter 1 overflow interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
17	SETOVL0INT	0	Set free running counter 0 overflow interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
16	SETTBINT	0	Set timebase interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
15-12	Reserved	0	Reads return 0. Writes have no effect.
11	SETDMA3	0	Set compare DMA request 3. <i>Read:</i> DMA request is disabled. <i>Write:</i> DMA request is unchanged.
		1	<i>Read or Write:</i> DMA request is enabled.
10	SETDMA2	0	Set compare DMA request 2. <i>Read:</i> DMA request is disabled. <i>Write:</i> DMA request is unchanged.
		1	<i>Read or Write:</i> DMA request is enabled.
9	SETDMA1	0	Set compare DMA request 1. <i>Read:</i> DMA request is disabled. <i>Write:</i> DMA request is unchanged.
		1	<i>Read or Write:</i> DMA request is enabled.

Table 13-26. RTI Set Interrupt Control Register (RTISETINTENA) Field Descriptions (continued)

Bit	Field	Value	Description
8	SETDMA0	0	Set compare DMA request 0. <i>Read:</i> DMA request is disabled. <i>Write:</i> DMA request is unchanged.
		1	<i>Read or Write:</i> DMA request is enabled.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3	SETINT3	0	Set compare interrupt 3. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
2	SETINT2	0	Set compare interrupt 2. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
1	SETINT1	0	Set compare interrupt 1. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.
0	SETINT0	0	Set compare interrupt 0. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read or Write:</i> Interrupt is enabled.

13.3.26 RTI Clear Interrupt Enable Register (RTICLEARINTENA)

This register prevents the necessity of a read-modify-write operation if a particular interrupt should be disabled. This register is shown in [Figure 13-37](#) and described in [Table 13-27](#).

Figure 13-37. RTI Clear Interrupt Control Register (RTICLEARINTENA) [offset = 84h]

31																	24
Reserved																	
R-0																	
23								19	18		17		16				
Reserved								CLEAROVL1INT		CLEAROVL0INT		CLEARTBINT					
R-0								R/WP-0		R/WP-0		R/WP-0					
15								12	11		10		9		8		
Reserved								CLEARDMA3		CLEARDMA2		CLEARDMA1		CLEARDMA0			
R-0								R/WP-0		R/WP-0		R/WP-0		R/WP-0			
7								4	3		2		1		0		
Reserved								CLEARINT3		CLEARINT2		CLEARINT1		CLEARINT0			
R-0								R/WP-0		R/WP-0		R/WP-0		R/WP-0			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-27. RTI Clear Interrupt Control Register (RTICLEARINTENA) Field Descriptions

Bit	Field	Value	Description
31-19	Reserved	0	Reads return 0. Writes have no effect.
18	CLEAROV1INT	0	Clear free running counter 1 overflow interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
17	CLEAROV0INT	0	Clear free running counter 0 overflow interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
16	CLEARTBINT	0	Clear timebase interrupt. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
15-12	Reserved	0	Reads return 0. Writes have no effect.
11	CLEARDMA3	0	Clear compare DMA request 3. <i>Read:</i> DMA request is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> DMA request is enabled. <i>Write:</i> DMA request is disabled.
10	CLEARDMA2	0	Clear compare DMA request 2. <i>Read:</i> DMA request is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> DMA request is enabled. <i>Write:</i> DMA request is disabled.

Table 13-27. RTI Clear Interrupt Control Register (RTICLEARINTENA) Field Descriptions (continued)

Bit	Field	Value	Description
9	CLEARDMA1	0	Clear compare DMA request 1. <i>Read:</i> DMA request is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> DMA request is enabled. <i>Write:</i> DMA request is disabled.
8	CLEARDMA0	0	Clear compare DMA request 0. <i>Read:</i> DMA request is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> DMA request is enabled. <i>Write:</i> DMA request is disabled.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3	CLEARINT3	0	Clear compare interrupt 3. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
2	CLEARINT2	0	Clear compare interrupt 2. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
1	CLEARINT1	0	Clear compare interrupt 1. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.
0	CLEARINT0	0	Clear compare interrupt 0. <i>Read:</i> Interrupt is disabled. <i>Write:</i> Corresponding bit is unchanged.
		1	<i>Read:</i> Interrupt is enabled. <i>Write:</i> Interrupt is disabled.

13.3.27 RTI Interrupt Flag Register (RTIINTFLAG)

The corresponding flags are set at every compare match of the RTIFRCx and RTICOMPx values, whether the interrupt is enabled or not. This register is shown in [Figure 13-38](#) and described in [Table 13-28](#).

Figure 13-38. RTI Interrupt Flag Register (RTIINTFLAG) [offset = 88h]

31		19	18	17	16
Reserved			OVL1INT	OVL0INT	TBINT
R-0			R/W1CP-0	R/W1CP-0	R/W1CP-0
15		4	3	2	1
Reserved			INT3	INT2	INT1
R-0			R/W1CP-0	R/W1CP-0	R/W1CP-0
					INT0
					R/W1CP-0

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 to clear in privilege mode only; -n = value after reset

Table 13-28. RTI Interrupt Flag Register (RTIINTFLAG) Field Descriptions

Bit	Field	Value	Description
31-19	Reserved	0	Reads return 0. Writes have no effect.
18	OVL1INT	0	Free running counter 1 overflow interrupt flag. This bit determines if an interrupt is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.
17	OVL0INT	0	Free running counter 0 overflow interrupt flag. This bit determines if an interrupt is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.
16	TBINT	0	Timebase interrupt flag. This flag is set when the TBEXT bit is cleared by detection of a missing external clock edge. It will not be set by clearing TBEXT by software. It determines if an interrupt is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.
15-4	Reserved	0	Reads return 0. Writes have no effect.
3	INT3	0	Interrupt flag 3. These bits determine if an interrupt due to a Compare 3 match is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.
2	INT2	0	Interrupt flag 2. These bits determine if an interrupt due to a Compare 2 match is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.
1	INT1	0	Interrupt flag 1. These bits determine if an interrupt due to a Compare 1 match is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.

Table 13-28. RTI Interrupt Flag Register (RTIINTFLAG) Field Descriptions (continued)

Bit	Field	Value	Description
0	INT0	0	Interrupt flag 0. These bits determine if an interrupt due to a Compare 0 match is pending. <i>Read:</i> No interrupt is pending. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Interrupt is pending. <i>Write:</i> Bit is cleared to 0.

13.3.28 Digital Watchdog Control Register (RTIDWDCTRL)

The software has to write to the DWDCTRL field in order to enable the DWD, as described below. Once enabled, the watchdog can only be disabled by a system reset. The application cannot disable the watchdog. However should the RTICLK source be changed to a source that is unimplemented it will have the same effect as disabling the watchdog. This register is shown in [Figure 13-38](#) and described in [Table 13-28](#).

Figure 13-39. Digital Watchdog Control Register (RTIDWDCTRL) [offset = 90h]

31				16
DWDCTRL				
R/WP-5312h				
15				0
DWDCTRL				
R/WP-ACEDh				

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

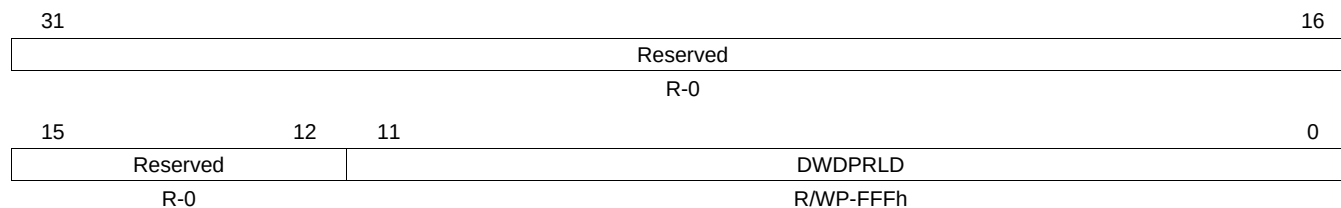
Table 13-29. Digital Watchdog Control Register (RTIDWDCTRL) Field Descriptions

Bit	Field	Value	Description
31-0	DWDCTRL	5312 ACEDh	Digital Watchdog Control. <i>Read:</i> DWD counter is disabled. <i>Write:</i> State of DWD counter is unchanged (stays enabled or disabled).
		A985 59DAh	<i>Read:</i> DWD counter is enabled. <i>Write:</i> DWD counter is enabled.
		All other values	<i>Read:</i> DWD counter state is unchanged (enabled or disabled). <i>Write:</i> State of DWD counter is unchanged (stays enabled or disabled). Note: Once the enable value is written, all other future writes are blocked. In other words, once DWD is enabled, it can only be disabled by system reset or power on reset. However should the RTICLK source be changed to a source that is unimplemented it will have the same effect as disabling the watchdog.

13.3.29 Digital Watchdog Preload Register (RTIDWDPRLD)

This register sets the expiration time of the DWD. This register is shown in [Figure 13-38](#) and described in [Table 13-28](#).

Figure 13-40. Digital Watchdog Preload Register (RTIDWDPRLD) [offset = 94h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-30. Digital Watchdog Preload Register (RTIDWDPRLD) Field Descriptions

Bit	Field	Value	Description
31-12	Reserved	0	Reads return 0 and writes have no effect.
11-0	DWDPRLD	0-FFFh	Digital Watchdog Preload Value. <i>Read:</i> The current preload value <i>Write:</i> Set the preload value. The DWD preload register can be configured only when the DWD is disabled. Therefore, the application can only configure the DWD preload register before it enables the DWD down counter. The expiration time of the DWD Down Counter can be determined with following equation: $t_{exp} = (DWDPRLD + 1) \times 2^{13} / RTICLK1$ where: DWDPRLD = 0...4095

13.3.30 Watchdog Status Register (RTIWDSTATUS)

This register records the status of the DWD. The values of the following status bits will not be affected by a soft reset. These bits are cleared by a power-on reset, or by a write of 1. These bits can be used for debug purposes. This register is shown in [Figure 13-38](#) and described in [Table 13-28](#).

Figure 13-41. Watchdog Status Register (RTIWDSTATUS) [offset = 98h]

31							8	
Reserved								
R-0								
7		6	5	4	3	2	1	0
Reserved		DWWD ST	END TIME VIOL	START TIME VIOL	KEY ST	DWD ST	Reserved	
R-0		R/W1CP-x	R/W1CP-x	R/W1CP-x	R/W1CP-x	R/W1CP-x	R/W1CP-x	R-0

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 to clear in privilege mode only; -n = value after reset

Table 13-31. Watchdog Status Register (RTIWDSTATUS) Field Descriptions

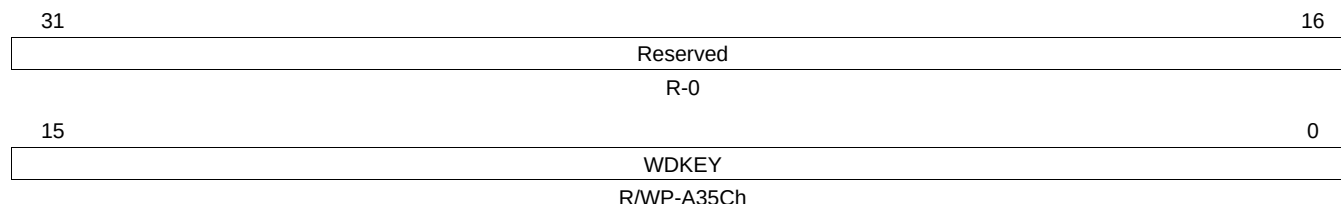
Bit	Field	Value	Description
31-6	Reserved	0	Reads return 0. Writes have no effect.
5	DWWD ST	0	Windowed Watchdog Status <i>Read:</i> No time-window violation has occurred. <i>Write:</i> Leaves the current value unchanged.
		1	<i>Read:</i> Time-window violation has occurred. The watchdog has generated either a system reset or a non-maskable interrupt to the CPU in this case. <i>Write:</i> Bit is cleared to 0. This will also clear all other status flags in the RTIWDSTATUS register. Clearing of the status flags will deassert the non-maskable interrupt generated due to violation of the DWWD.
4	END TIME VIOL	0	Windowed Watchdog End Time Violation Status. This bit indicates whether the Watchdog counter expired. <i>Read:</i> No end-time window violation has occurred. <i>Write:</i> Leaves the current value unchanged.
		1	<i>Read:</i> End-time defined by the windowed watchdog configuration has been violated. <i>Write:</i> Bit is cleared to 0.
3	START TIME VIOL	0	Windowed Watchdog Start Time Violation Status. This bit indicates whether the key is written before the watchdog window opened up. <i>Read:</i> No start-time window violation has occurred. <i>Write:</i> Leaves the current value unchanged.
		1	<i>Read:</i> Start-time defined by the windowed watchdog configuration has been violated. <i>Write:</i> Bit is cleared to 0.
2	KEY ST	0	Watchdog key status. This bit indicates a reset or NMI generated by a wrong key or key sequence written to the RTIWDKEY register. <i>Read:</i> No wrong key or key-sequence written. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Wrong key or key-sequence written to RTIWDKEY register. <i>Write:</i> Bit is cleared to 0.
1	DWD ST	0	DWD status. This bit is equivalent to bit END TIME VIOL. <i>Read:</i> No reset or NMI was generated. <i>Write:</i> Bit is unchanged.
		1	<i>Read:</i> Reset or NMI was generated. <i>Write:</i> Bit is cleared to 0.
0	Reserved	0	Reads return 0. Writes have no effect.

13.3.31 RTI Watchdog Key Register (RTIWDKEY)

This register must be written with the correct written key values to serve the watchdog. This register is shown in [Figure 13-42](#) and described in [Table 13-32](#).

NOTE: It has to be taken into account that the write to the RTIWDKEY register takes 3 VCLK cycles.

Figure 13-42. RTI Watchdog Key Register (RTIDWDKEY) [offset = 9Ch]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-32. RTI Watchdog Key Register (RTIDWDKEY) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0 and writes have no effect.
15-0	WDKEY	0-FFFFh	Watchdog key. These bits provide the key sequence location. Reads returns the current WDKEY value. A write of E51Ah followed by A35Ch in two separate write operations defines the key sequence and reloads the DWD. Writing any other value causes a reset or NMI, as shown in Table 13-33 . Writing any other value will cause the WDKEY to reset to A35Ch.

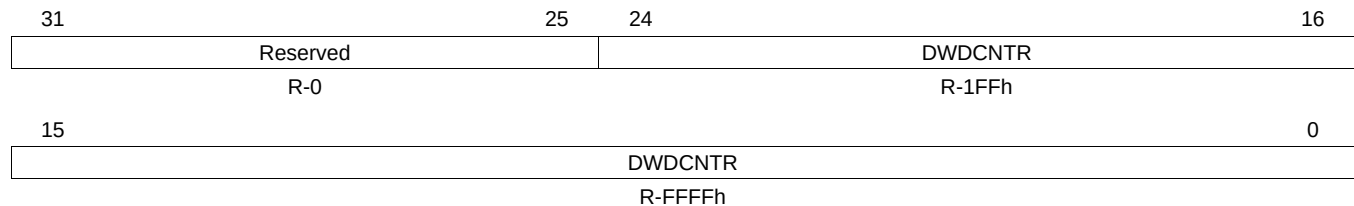
Table 13-33. Example of a WDKEY Sequence

Step	Value Written to WDKEY	Result
1	A35Ch	No action
2	A35Ch	No action
3	E51Ah	WDKEY is enabled for reset or NMI by next A35Ch.
4	E51Ah	WDKEY is enabled for reset or NMI by next A35Ch.
5	E51Ah	WDKEY is enabled for reset or NMI by next A35Ch.
6	A35Ch	Watchdog is reset.
7	A35Ch	No action
8	E51Ah	WDKEY is enabled for reset or NMI by next A35Ch.
9	A35Ch	Watchdog is reset.
10	E51Ah	WDKEY is enabled for reset or NMI by next A35Ch.
11	2345h	System reset or NMI; incorrect value written to WDKEY.

13.3.32 RTI Digital Watchdog Down Counter (RTIDWDCNTR)

This register provides the current value of the DWD down counter. This register is shown in [Figure 13-43](#) and described in [Table 13-34](#).

Figure 13-43. RTI Watchdog Down Counter Register (RTIDWDCNTR) [offset = A0h]



LEGEND: R = Read only; -n = value after reset

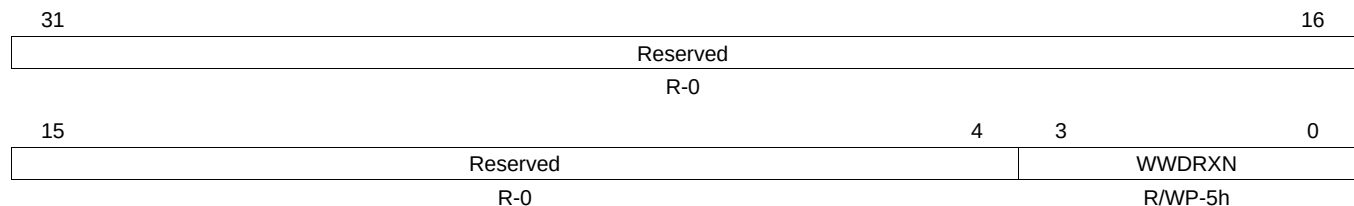
Table 13-34. RTI Watchdog Down Counter Register (RTIDWDCNTR) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0 and writes have no effect.
24-0	DWD CNTR	0-1FF FFFFh	DWD down counter. Reads return the current counter value.

13.3.33 Digital Windowed Watchdog Reaction Control (RTIWWDRXNCTRL)

This register selects the DWWD reaction if the watchdog is serviced outside the time window. This register is shown in [Figure 13-44](#) and described in [Table 13-35](#).

Figure 13-44. Digital Windowed Watchdog Reaction Control (RTIWWDRXNCTRL) [offset = A4h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-35. Digital Windowed Watchdog Reaction Control (RTIWWDRXNCTRL) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0 and writes have no effect.
3-0	WWDRXN		The DWWD reaction
		5h	The windowed watchdog will cause a reset if the watchdog is serviced outside the time window defined by the configuration, or if the watchdog is not serviced at all.
		Ah	The windowed watchdog will generate a non-maskable interrupt to the CPU if the watchdog is serviced outside the time window defined by the configuration, or if the watchdog is not serviced at all.
		All other values	The windowed watchdog will cause a reset if the watchdog is serviced outside the time window defined by the configuration, or if the watchdog is not serviced at all. Note: The DWWD reaction can be selected by the application even when the DWWD counter is already enabled. If a change to the WWDRXN is made before the watchdog service window is opened, then the change in the configuration takes effect immediately. If a change to the WWDRXN is made when the watchdog service window is already open, then the change in configuration takes effect only after the watchdog is serviced.

13.3.34 Digital Windowed Watchdog Window Size Control (RTIWWDSIZECTRL)

This register selects the DWWD window size. This register is shown in [Figure 13-45](#) and described in [Table 13-36](#).

Figure 13-45. Digital Windowed Watchdog Window Size Control (RTIWWDSIZECTRL) [offset = A8h]

31	16
WWDSIZE	
R/WP-0000	
15	0
WWDSIZE	
R/WP-0005h	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

**Table 13-36. Digital Windowed Watchdog Window Size Control (RTIWWDSIZECTRL)
Field Descriptions**

Bit	Field	Value	Description
31-0	WWDSIZE	0	The DWWD window size
		0000 0005h	100% (The functionality is the same as the standard time-out digital watchdog.)
		0000 0050h	50%
		0000 0500h	25%
		0000 5000h	12.5%
		0005 0000h	6.25%
		All other values	3.125%
			Note: The DWWD window size can be selected by the application even when the DWWD counter is already enabled. If a change to the WWDSIZE is made before the watchdog service window is opened, then the change in the configuration takes effect immediately. If a change to the WWDSIZE is made when the watchdog service window is already open, then the change in configuration takes effect only after the watchdog is serviced.

13.3.35 RTI Compare Interrupt Clear Enable Register (RTIINTCLREnable)

When the RTI compare event is configured to generate a DMA request or triggers (all triggered by RTI compare interrupt request flag) to other peripherals, it is often desirable to clear the RTI compare flag automatically so that the requests can be generated repeatedly without any CPU intervention. This register works with the RTI compare clear registers to enable an "auto-clear" of the compare interrupt enable bit after a compare equal event. This register is shown in Figure 13-46 and described in Table 13-37.

Figure 13-46. RTI Compare Interrupt Clear Enable Register (RTIINTCLREnable) [offset = ACh]

31	28	27	24	23	20	19	16
Reserved				INTCLREnable3			
R-0				R/WP-5h			
15	12	11	8	7	4	3	0
Reserved				INTCLREnable1			
R-0				R/WP-5h			
Reserved				INTCLREnable0			
R-0				R/WP-5h			

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 13-37. RTI Compare Interrupt Clear Enable Register (RTIINTCLREnable) Field Descriptions

Bit	Field	Value	Description
31-28	Reserved	0	Reads return 0. Writes have no effect.
27-24	INTCLREnable3	5h	Enables the auto-clear functionality on the compare 3 interrupt. <i>Read:</i> Auto-clear for compare 3 interrupt is disabled. <i>Privileged Write:</i> Auto-clear for compare 3 interrupt becomes disabled.
		All other values	<i>Read:</i> Auto-clear for compare 3 interrupt is enabled. <i>Privileged Write:</i> Auto-clear for compare 3 interrupt becomes enabled.
23-20	Reserved	0	Reads return 0. Writes have no effect.
19-16	INTCLREnable2	5h	Enables the auto-clear functionality on the compare 2 interrupt. <i>Read:</i> Auto-clear for compare 2 interrupt is disabled. <i>Privileged Write:</i> Auto-clear for compare 2 interrupt becomes disabled.
		All other values	<i>Read:</i> Auto-clear for compare 2 interrupt is enabled. <i>Privileged Write:</i> Auto-clear for compare 2 interrupt becomes enabled.
15-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	INTCLREnable1	5h	Enables the auto-clear functionality on the compare 1 interrupt. <i>Read:</i> Auto-clear for compare 1 interrupt is disabled. <i>Privileged Write:</i> Auto-clear for compare 1 interrupt becomes disabled.
		All other values	<i>Read:</i> Auto-clear for compare 1 interrupt is enabled. <i>Privileged Write:</i> Auto-clear for compare 1 interrupt becomes enabled.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	INTCLREnable0	5h	Enables the auto-clear functionality on the compare 0 interrupt. <i>Read:</i> Auto-clear for compare 0 interrupt is disabled. <i>Privileged Write:</i> Auto-clear for compare 0 interrupt becomes disabled.
		All other values	<i>Read:</i> Auto-clear for compare 0 interrupt is enabled. <i>Privileged Write:</i> Auto-clear for compare 0 interrupt becomes enabled.

13.3.36 RTI Compare 0 Clear Register (RTICMP0CLR)

This registers holds an initial value which is larger than the value in the RTI Compare 0 register [Section 13.3.4](#). The user needs to choose the value such that the compare clear 0 event occurs before next compare 0 event. If the Free Running Counter matches the compare value, the compare 0 interrupt request flag is cleared and the value in the RTIUDCP0 register [Section 13.3.16](#) is added to this register. This register is shown in [Figure 13-47](#) and described in [Table 13-38](#).

Figure 13-47. RTI Compare 0 Clear Register (RTICMP0CLR) [offset = B0h]

31	16
CMP0CLR	
R/WP-0	
15	0
CMP0CLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-38. RTI Compare 0 Clear Register (RTICMP0CLR) Field Descriptions

Bit	Field	Value	Description
31-0	CMP0CLR	0-FFFF FFFFh	Compare 0 clear. This registers holds a compare value. If the Free Running Counter matches the compare value, the compare 0 interrupt request flag is cleared and the value in the RTIUDCP0 register Section 13.3.16 is added to this register. Reads return the current compare clear value. A privileged write to this register updates the compare clear value.

13.3.37 RTI Compare 1 Clear Register (RTICMP1CLR)

This registers holds an initial value which is larger than the value in the RTI Compare 1 register [Section 13.3.4](#). The user needs to choose the value such that the compare clear 1 event occurs before next compare 1 event. If the Free Running Counter matches the compare value, the compare 1 interrupt request flag is cleared and the value in the RTIUDCP1 register [Section 13.3.18](#) is added to this register. This register is shown in [Figure 13-48](#) and described in [Table 13-39](#).

Figure 13-48. RTI Compare 1 Clear Register (RTICMP1CLR) [offset = B4h]

31	16
CMP1CLR	
R/WP-0	
15	0
CMP1CLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

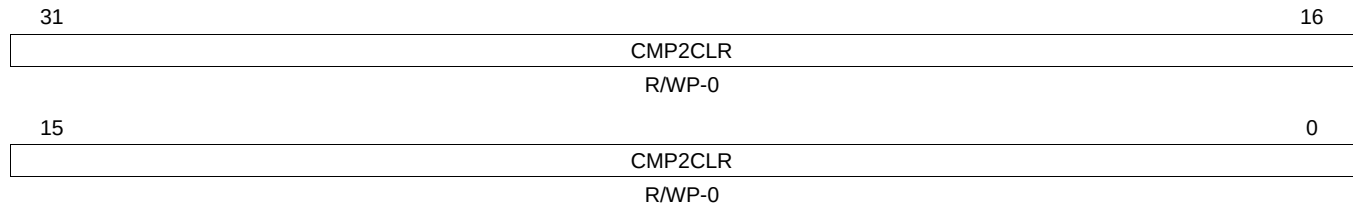
Table 13-39. RTI Compare 1 Clear Register (RTICMP1CLR) Field Descriptions

Bit	Field	Value	Description
31-0	CMP0CLR	0-FFFF FFFFh	Compare 1 clear. This registers holds a compare value. If the Free Running Counter matches the compare value, the compare 1 interrupt request flag is cleared and the value in the RTIUDCP1 register Section 13.3.18 is added to this register. Reads return the current compare clear value. A privileged write to this register updates the compare clear value.

13.3.38 RTI Compare 2 Clear Register (RTICMP2CLR)

This registers holds an initial value which is larger than the value in the RTI Compare 2 register [Section 13.3.4](#). The user needs to choose the value such that the compare clear 2 event occurs before next compare 2 event. If the Free Running Counter matches the compare value, the compare 2 interrupt request flag is cleared and the value in the RTIUDCP2 register [Section 13.3.20](#) is added to this register. This register is shown in [Figure 13-49](#) and described in [Table 13-40](#).

Figure 13-49. RTI Compare 2 Clear Register (RTICMP2CLR) [offset = B8h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

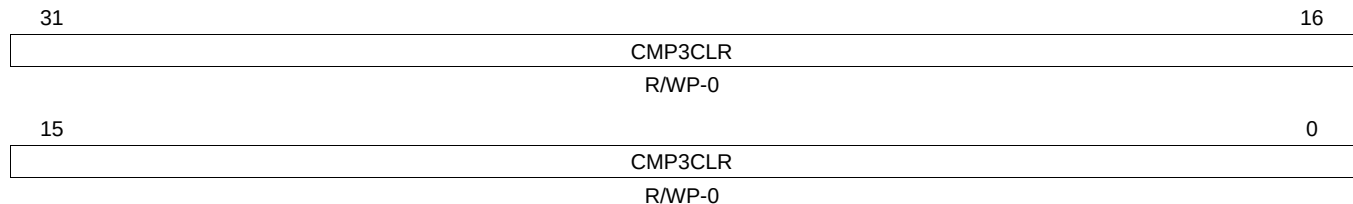
Table 13-40. RTI Compare 2 Clear Register (RTICMP2CLR) Field Descriptions

Bit	Field	Value	Description
31-0	CMP2CLR	0-FFFF FFFFh	Compare 2 clear. This registers holds a compare value. If the Free Running Counter matches the compare value, the compare 2 interrupt request flag is cleared and the value in the RTIUDCP2 register Section 13.3.20 is added to this register. Reads return the current compare clear value. A privileged write to this register updates the compare clear value.

13.3.39 RTI Compare 3 Clear Register (RTICMP3CLR)

This registers holds an initial value which is larger than the value in the RTI Compare 3 register [Section 13.3.4](#). The user needs to choose the value such that the compare clear 3 event occurs before next compare 3 event. If the Free Running Counter matches the compare value, the compare 3 interrupt request flag is cleared and the value in the RTIUDCP3 register [Section 13.3.22](#) is added to this register. This register is shown in [Figure 13-50](#) and described in [Table 13-41](#).

Figure 13-50. RTI Compare 3 Clear Register (RTICMP3CLR) [offset = BCh]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 13-41. RTI Compare 3 Clear Register (RTICMP3CLR) Field Descriptions

Bit	Field	Value	Description
31-0	CMP3CLR	0-FFFF FFFFh	Compare 3 clear. This registers holds a compare value. If the Free Running Counter matches the compare value, the compare 3 interrupt request flag is cleared and the value in the RTIUDCP3 register Section 13.3.22 is added to this register. Reads return the current compare clear value. A privileged write to this register updates the compare clear value.